

Chapter 15

Radiosurgery for Uveal Melanoma

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Abstract Uveal melanomas are the most common primary intraocular malignancy. In patients with localized disease eye preservation is often sought if feasible. Radiation therapy has been used as a local organ preserving treatment for decades. Conventional radiation therapy techniques includes plaque brachy therapy and proton beam radiation. While these are effective, they can be invasive or expensive with limited availability. Stereotactic radiosurgery (SRS) and radiotherapy is an evolving modality to deliver conformal radiation with the intent towards eye preservation.

This Chapter outlines the role SRS in the treatment of uveal melanomas in the setting of a large single institution experience with a practical guide on patient selection, treatment planning and outcomes assessment.

Objective: To analyze the local efficacy and eye retention rate after frameless, image-guided robotic radiosurgery against uveal melanoma.

Methods: 200 patients with mainly medium sized and large unilateral uveal melanomas (3 % small, 62 % medium, 35 % large) were treated with a frameless robotic radiosurgery system. Median age was 61 years (range 32–78 years). All patients underwent a single-session procedure beginning with retrobulbar anaesthesia, followed by magnetic resonance imaging (MRI) and computerized tomography (CT) scanning that was used for generation of the treatment plan. The tumor dose was between 18 and 22 Gy (mean: 21 Gy) prescribed to the 70 % isodose line. Three-dimensional treatment planning was aimed at securing the optical lens and the optic disc as much as possible. Follow-up occurred at 3, 6, 12, 18 months and yearly thereafter with clinical, ultrasound and MRI studies.

Results: The median follow up time was 18 months. All patients could be treated in the frameless setup within 3 h. The local control after 1, 2 and 4 years was 97.7,

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92.4 and 77.7 %, respectively. Nineteen of the 200 patients (9.9 %) underwent enucleation because of tumor growth and/or complications (glaucoma, tumor bleeding) during the follow up period.

Conclusion: Frameless, single-session, image-guided robotic radiosurgery is an effective and straight forward treatment option for patients with medium sized and large uveal melanoma which are otherwise difficult to treat.

Keywords Uveal Melanoma • Cyberknife • Radiosurgery • Frameless

15.1 Introduction

Uveal melanomas comprise of 70 % of all primary intraocular malignancies [1]. The incidence is 6–7 in one million (98.7 % white) population with an overall risk for metastasis of 50 % depending on patient age, tumor size and incidence of subretinal fluid, hemorrhage, and extraocular extension [2]. Patients with metastasis are living longer due to the benefit of various treatment approaches like partial hepatectomy; radiofrequency ablation; ipilimumab immunotherapy; selective internal radiotherapy; intra-hepatic chemotherapy, possibly with isolated liver perfusion; and systemic chemotherapy. The 5-year survival rate is 80 % after surgery or radiotherapy.

The most common manifestation of uveal melanoma is in the choroid (85 %) followed by the ciliary body (10 %) and the iris (5 %). Although genetic factors like loss of heterozygosity (LOH) of chromosome 3 are known to increase the risk of metastasis significantly, there is no familial clustering [3]. Therefore, uveal melanoma is not regarded as an inherited disease and displays an equal sex distribution. In most cases, only one eye is affected in patients older than 50 years of age. It affects in 97.8 % the white population.

Different forms of radiation therapy are now used against uveal melanoma, replacing enucleation as the treatment of choice. Surgery is still used for tumors of large size (prominence over 10 mm) Fractionated proton radiation therapy, radioactive eye plaques, conventional LINAC radiotherapy and frame-based radiosurgical techniques have been described and shown to be effective for local tumor control [4–27].

For small and medium-sized uveal melanoma, ophthalmic plaque radiotherapy is a safe and effective treatment using ruthenium 106 (most cases; beta irradiation), iodine 125 (photon emission) or palladium 103 (photon emission) as radioactive isotopes for therapy. However, for precise placement of the plaque, eye muscles often have to be detached temporarily and the patient needs to stay isolated in the hospital for several days depending on the duration of brachytherapy. Alternative treatment methods include transpupillary thermotherapy and photodynamic therapy which can be considered for small-size melanoma located at the posterior pole close to the optic nerve or the macula. However, longterm results have not been convincing for TTT or PDT as a single treatment approach, but in combination with plaque brachytherapy they might be promising [28].

Single-session radiosurgery using frame-based Gamma Knife® (Elekta AB, Stockholm, Sweden) technology has been used during the last decade to treat middle- to large-size tumors as an alternative to complete removal of the eye [22–24, 29, 30]. Different methods of eye-tracking and diverse fixation techniques, along with the stereotactic head frame application have been described to keep the eye immobile for the duration of treatment [30]. These difficult, and for the patient, highly bothersome techniques challenge somewhat the definition of “minimally invasive treatment”.

Today, with the availability of frameless robotic radiosurgery technology, we are able to apply a treatment modality without any eye fixation or frame application. Using local retrobulbar anaesthesia and an image-guided robotic radiosurgery system, a minimally invasive radiosurgical treatment can be delivered [31, 32].

15.2 Indications for SBRT Treatment (I.E Case Selection and Exclusions)

SRS or SBRT is used predominantly for tumors of medium and large size not suitable for brachytherapy (tumor height >6 mm, tumor base >19 mm). Recently also smaller tumors underwent SRS in our centre for patients not willing to undergo brachytherapy which requires surgical intervention with implantation and explantation of the radioactive plaque.

15.3 Radiosurgery Treatment Process

Radiosurgery is an alternative method when the size and location of the tumor is not amenable for brachytherapy or because of the patients’ wish to avoid primary enucleation. Table 15.1 shows the main treatment techniques for uveal melanoma. In selected cases patients may undergo radiosurgery after prior unsuccessful brachytherapy to the target lesion.

All patients need to be evaluated by an ophthalmologist and a specialized radiation oncologist for eligibility of treatment. The following standard outpatient procedure we developed in our centre is recommended:

Table 15.1 Treatment options uveal melanoma

Small melanoma (<6 mm)	Medium/large melanoma (≥6 mm)
Brachytherapy 106 Ru	Eye enucleation
Transpupillary thermotherapy	Cyberknife radiosurgery
Proton beam therapy	Gamma Knife radiosurgery
	Brachytherapy 125 iodine
	Proton beam, helium ions therapy

Patients are positioned on the computed tomography (CT) couch and standard retrobulbar anesthesia with goal of complete akinesia of the globe within the orbit is applied. The volume of anesthesia is dependent on the volume of the orbit and is typically between 10 and 15 ml. Suturing of the rectus muscles as described seems not to be needed anymore. A contrast-enhanced CT of the head with 1-mm slice thickness and a 1 mm T2 MRI of the orbit is performed after retrobulbar anesthesia.

Based on the fused CT and MRI sequences, the target volume is defined by the ophthalmologist and the radiosurgeon, and the treatment plan is generated. In all cases, the tumor is covered completely by the 70 % isodose line. We add a 1 mm margin to the target volume to all directions and 2 mm to the posterior tumor border (Fig. dose plan uvea). This is done to compensate for possible posterior movement of the eye bulb after retrobulbar anesthesia. After the planning procedure, the treatment plan is loaded for treatment delivery. Single-session radiosurgery follows immediately after the plan generation is finalized. Radiation is delivered in a single fraction for all cases using 18–22 Gy to the prescription isodose (70 % in the case of Cyberknife) depending on the size and location of the tumor. After radiosurgery the patient is discharged. Typically there are no significant acute toxicities to be expected. Depending on the size and location of the tumor late side effects can develop several weeks or months after treatment with rubeosis, glaucoma or cataract. A secondary glaucoma is the late toxicity most often encountered. About 25 % of patients will develop a secondary glaucoma 3–60 months (mean 21 months after SRS). These side effects are highly individual and also depend on the individual radiation sensitivity of the patient.

Clinical and imaging follow-up will be performed 3, 6, 12 and 18 months after the treatment using standardized A- and B-scan ultrasound and a 1 mm T2 MRI for evaluation of local control. Tumor control is defined as either continuous regression of the tumor or no further progression.

15.4 Results

We used this method to treat 200 patients with uveal melanoma. The mean follow-up for these patients was 18 months. All patients were treated in one treatment session comprising of retrobulbar anaesthesia, CT and MRI imaging, treatment planning and treatment application within 3 h after start of the anaesthesia. The local control after 1, 2 and 4 years was 97.7, 92.4 and 77.7 %, respectively. Treatment parameters as well as dose to the lens of the affected and contralateral eye and point dose to the optic disc of our first 20 Cyberknife patients are shown in Table 15.2.

The median maximum apical tumor height according to standardized A-scan ultrasound before treatment was 8 mm (mean 7.9 mm $\pm$ 2.6 mm) compared to 5 mm (mean 5.4 $\pm$ 2.1 mm) at the last follow-up examination ($p < 0.1$) (Figs. 15.1 and 15.2). The visual acuity decreased in most patients over time, typically between 6

Table 15.2 Dose parameters first 20 patients of the Munich series

Patient nr.	Dose	Isodose	Dmax lens affected eye (Gy)	Dmax lens contralateral eye (Gy)	Point dose optic disk affected eye (Gy)
1	20	75	4.8	0	19.7
2	18	70	20.9	1.2	3.5
3	20	70	1.7	0.2	27.3
4	20	75	24.3	0	18.3
5	18	65	20.3	0	4.7
6	22	70	1.7	1.7	10
7	19	70	20.3	0	6
8	21	70	1.2	0.3	13.1
9	18	60	13.8	0	17.2
10	17	70	20.3	0.2	8.1
11	19	70	4.5	0.3	23.2
12	18	70	1.6	0.8	14.9
13	18	70	2.5	0.6	12.3
14	18	70	9.1	1.2	22.6
15	20	70	Blind	0.3	Blind
16	20	70	24.5	0.4	3.3
17	19	70	7.8	0.4	3.7
18	18	70	1.9	0.6	23.4
19	18	70	11.3	0.2	4.3
20	20	70	10.1	0.6	23.6

and 24 months after radiosurgery. The median reflectivity significantly increased during the follow-up period from 42 to 71 % ($p < 0.01$). Nineteen patients needed enucleation due to tumor re-growth after radiosurgery (9.5 %). Two patients were successfully retreated by radiosurgery.

15.5 Discussion

Different kinds of therapeutic options exist today for the treatment of uveal melanoma. However, enucleation has remained the gold standard therapy for the condition for many years [19]. Eye conservation is achieved by several techniques, with proton or other heavy charged particle therapies, and episcleral radionuclide plaque therapy being among the most commonly used [7–11]. Adams et al., for example, found no statistically significant survival difference in 223 patients treated with brachytherapy compared to 416 patients who underwent enucleation [4]. Useful vision is usually preserved mainly in cases in which tumors are located in a relatively favorable location with respect to the optic disc or macula. Stereotactic radiosurgery (SRS) and fractionated stereotactic radiotherapy (FSRT) are gaining wider acceptance in the medical community as the clinical data mature, and published evidence

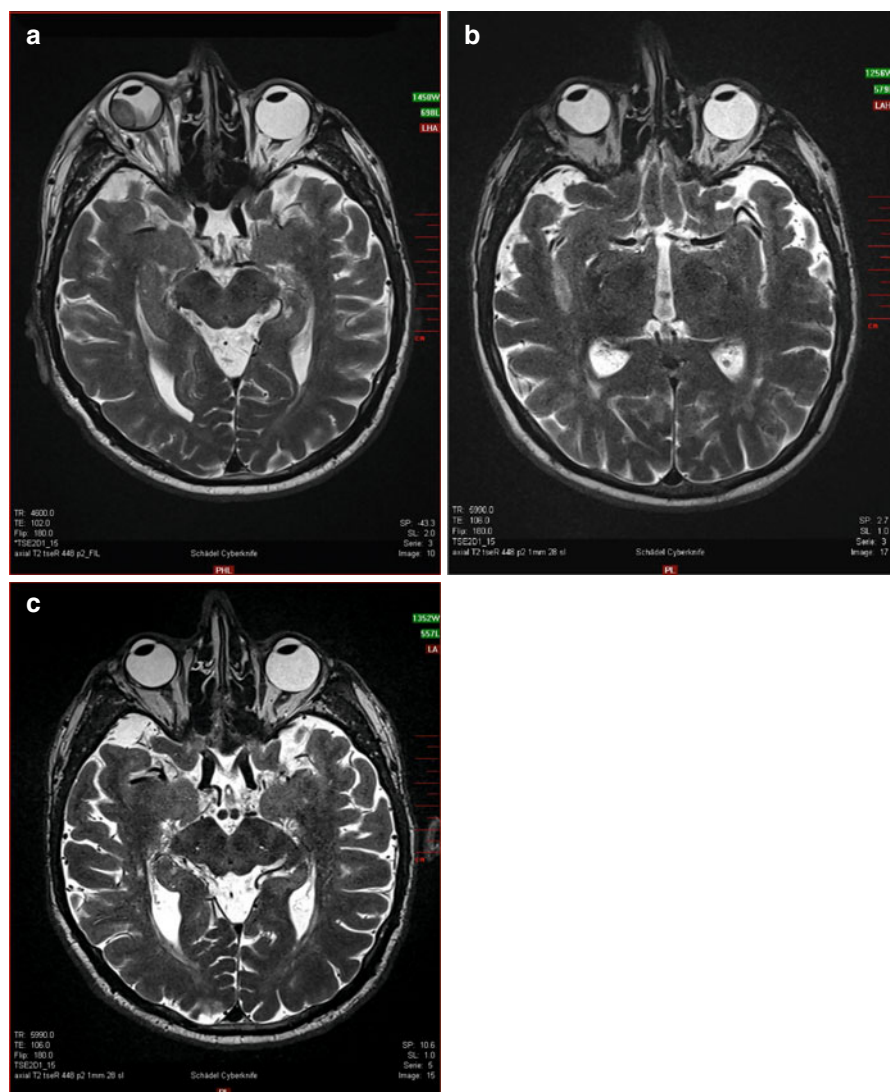


Fig. 15.1 (a) T2 MRI sequence showing large Uveal Melanoma on the temporal right eye with associated retinal detachment. (b) Follow up after 9 months shows significant tumor volume reduction and reversible retinal detachment. (c) Follow up after 18 months displays complete regression of the melanoma

shows this non-invasive treatment option to be reasonably safe and effective [1, 12–19] SRS requires immobilization of the eye in order to accurately plan and deliver the high-dose treatment. Most centers use a stereotactic frame (frame-based stereotactic radiosurgery), retrobulbar anesthesia and suturing of 2–4 rectus muscles. Others have described suction fixation devices for radiosurgical ocular treatments [29].

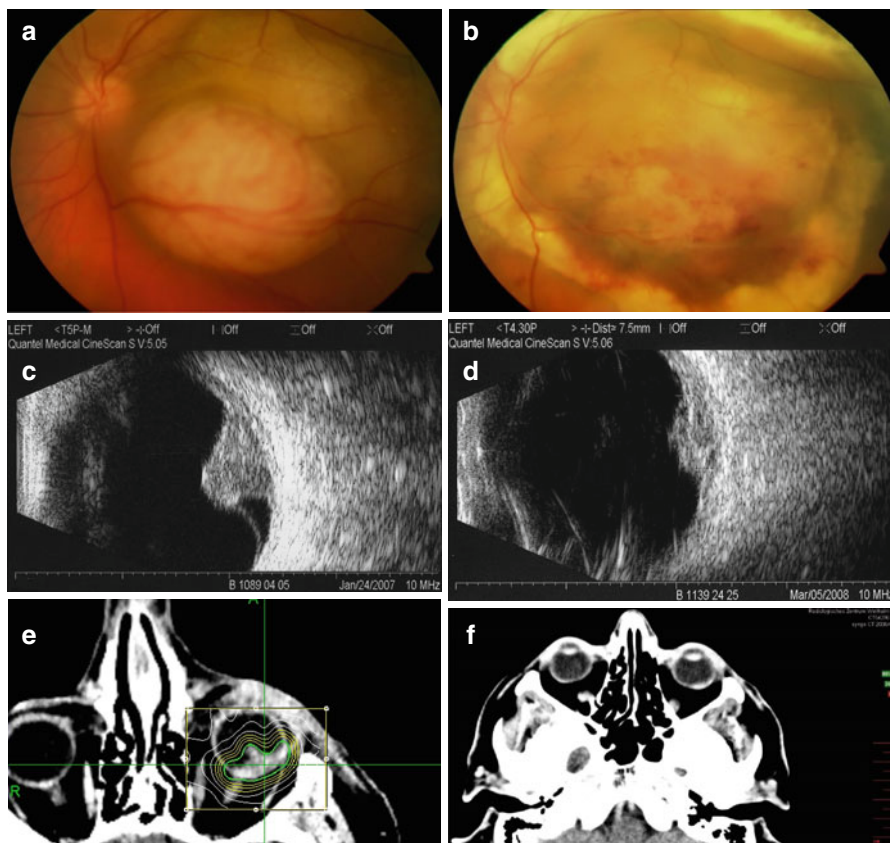


Fig. 15.2 (a, b) Funduscopy image before (a) and after (b) Cyberknife radiosurgery. (c, d) Ultrasound scan before (c) and after (d) Cyberknife radiosurgery. (e) Conformal plan with a steep dose gradient for a typical left-sided uveal melanoma. The isodoses from 10 to 70 % (prescribed isodose, *thick line*) are shown. The tumor was treated with single-session robotic radiosurgery procedure using a dose of 19 Gy. (f) Treatment result after 18 months. The tumor was significantly reduced in size

In our previous studies, akinesia of the eye after retrobulbar anesthesia and the induced proptosis lasted for at least 3 h in all of more than 200 treated cases, as shown by MRI studies before and immediately after the radiosurgical procedure [23, 24]. There was no significant soft tissue volume change during this time frame. Therefore, we concluded that retrobulbar anesthesia alone was sufficient for a time frame of at least 3 h and that no other additional eye-bulb immobilization devices were needed. This finding opened the possibility of carrying out the radiosurgical procedure without the highly invasive step of eye-immobilization and led us to develop the sequence of procedures described in this chapter.

We describe here a novel and minimally invasive treatment paradigm using frameless robotic SRS to treat uveal melanoma of the eye. We demonstrated that this treatment option was feasible for treating mid-size and large uveal melanomas

[31]. Compared with our previous series using frame-based SRS, it is far more non-invasive, safe, appears equally effective, and is considerably more comfortable for the patient. The only slightly invasive procedure left in the sequence is the administration of local anesthesia into the retrobulbar space, which is a safe and straightforward procedure routinely performed by experienced ophthalmologists [31]. Given the high tracking precision that retrobulbar anesthesia enables, we do not think that eye ball tracking techniques should replace this minimal invasive procedure with an inherent risk of greater inaccuracies during treatment.

In our earlier series, in which we treated medium- to large-size uveal melanomas, although the treatment proved effective, we did encounter certain adverse effects such as exudative retinal detachment in 20 % of the cases, and subsequently worse visual acuity [23, 24]. Generally, major treatment-related toxicities included neovascular glaucoma, radiation retinopathy or optic neuropathy, which were seen especially in the earlier series of Gamma Knife-based SRS, largely due to higher doses, larger tumor sizes and tumor locations [26, 27]. Larger basal diameter (>10 mm), ciliary body melanomas and doses over 35 Gy appeared to be associated with increased complication rates [17, 18]. Compared with eye-plaque therapy, Stereotactic radiosurgery comes with the advantage of not involving invasive surgery for plaque implantation and removal as well as hospitalization for several days required for radiation protection during brachytherapy. Fractionated proton therapy was not indicated for our patient cohort as proton centers are scarce and proton treatment costs are considerably higher than brachytherapy or radiosurgical treatment options.

Recently, there has been a trend towards dose de-escalation in this application. Several series showed equivalent local control but reduced toxicity rates with lower doses [22, 23, 26]. The dose level applied in the current study was lower than the earlier series. The treatment regimen was developed based on our previous experience and our ability to deliver dose plans with steeper gradients and better tumor coverage compared to the earlier, frame-based techniques we used.

The probability of visual preservation and eye retention with either technique are strongly dependent on tumor size and location. The dose to the lens and the optical disc is determined by the location of the tumor. However, for tumors of the lateral and posterior-lateral parts of the bulb, the dose to the lens and the optical disc could be kept to a minimum due to the steep dose gradient achieved using an inverse planning algorithm. Hirasawa et al. identified the anterior segment of the eye and the optic disc as structures of great risk for neovascular glaucoma based on a multivariate analysis, and recommended irradiation techniques that would spare these structures as much as possible when treating uveal melanoma [17]. The high flexibility of the robotic technology used in the current study is capable of achieving this goal [20]. Robotic radiosurgery has the ability to adjust the dose to the tumor as much as possible by maximally sparing the dose to sensitive structures of the eye.

Accurate treatment planning and delivery requires a stringent setup with an experienced interdisciplinary team comprised of ophthalmologists, radiation oncologists and imaging experts. Under optimal conditions, the described radiosurgical treatment paradigm seems to be safe, effective and comfortable for the patient.

15.6 Conclusions

Robotic radiosurgery is a safe and effective treatment approach for definitive treatment of medium sized and large uveal melanomas. MRI and CT image fusion for treatment planning enhance the exact tumor delineation. In experienced hands, retrobulbar anaesthesia, imaging, treatment planning and treatment application can be performed sequentially in one treatment session within a time frame of less than 3 h.

References

1. Henderson MA, Shirazi H, Lo SS, Mendonca MS, Fakiris AJ, Witt TC, Worth RM, Timmerman RD. Stereotactic radiosurgery and fractionated stereotactic radiotherapy in the treatment of uveal melanoma. *Technol Cancer Res Treat*. 2006;5:411–9.
2. Shields CL, Kaliki S, Furuta M, Fulco E, Alarcon C, Shields JA. American Joint Committee on Cancer Classification of posterior uveal melanoma (tumor size category) predicts prognosis in 7731 patients. *Ophthalmology*. 2013;120(10):2066–71.
3. Lake SL, Coupland SE, Taktak AF, Damato BE. Whole-genome microarray detects deletions and loss of heterozygosity of chromosome 3 occurring exclusively in metastasizing uveal melanoma. *Invest Ophthalmol*. 2010;51:4884–91.
4. Adams KS, Abramson DH, Ellsworth RM, Haik BG, Bedford M, Packer S, Seddon J, Albert D, Polivogianis L. Cobalt plaque versus enucleation for uveal melanoma: comparison of survival rates. *Br J Ophthalmol*. 1988;72:494–7.
5. Adler Jr JR, Chang SD, Murphy MJ, Doty J, Geis P, Hancock SL. The Cyberknife: a frameless robotic system for radiosurgery. *Stereotact Funct Neurosurg*. 1997;69:124–8.
6. Adler Jr JR, Murphy MJ, Chang SD, Hancock SL. Image-guided robotic radiosurgery. *Neurosurgery*. 1999;44:1299–306; discussion 1306–1297.
7. Augsburger JJ, Gamel JW, Lauritzen K, Brady LW. Cobalt-60 plaque radiotherapy vs enucleation for posterior uveal melanoma. *Am J Ophthalmol*. 1990;109:585–92.
8. Boyd SR, Gittos A, Richter M, Hungerford JL, Errington RD, Cree IA. Proton beam therapy and iris neovascularisation in uveal melanoma. *Eye (Lond)*. 2006;20:832–6.
9. Char DH, Bove R, Phillips TL. Laser and proton radiation to reduce uveal melanoma-associated exudative retinal detachments. *Am J Ophthalmol*. 2003;136:180–2.
10. Char DH, Phillips T, Daftari I. Proton teletherapy of uveal melanoma. *Int Ophthalmol Clin*. 2006;46:41–9.
11. Dendale R, Lumbroso-Le Rouic L, Noel G, Feuvret L, Levy C, Delacroix S, Meyer A, Nauraye C, Mazal A, Mammar H, Garcia P, D'Hermies F, Frau E, Plancher C, Asselain B, Schlienger P, Mazon JJ, Desjardins L. Proton beam radiotherapy for uveal melanoma: results of Curie Institut-Orsay proton therapy center (ICPO). *Int J Radiat Oncol Biol Phys*. 2006;65:780–7.
12. Dieckmann K, Bogner J, Georg D, Zehetmayer M, Kren G, Potter R. A linac-based stereotactic irradiation technique of uveal melanoma. *Radiother Oncol*. 2001;61:49–56.
13. Dieckmann K, Georg D, Zehetmayer M, Bogner J, Georgopoulos M, Potter R. LINAC based stereotactic radiotherapy of uveal melanoma: 4 years clinical experience. *Radiother Oncol*. 2003;67:199–206.
14. Dieckmann K, Georg D, Zehetmayer M, Rottenfusser A, Potter R. Stereotactic photon beam irradiation of uveal melanoma: indications and experience at the University of Vienna since 1997. *Strahlenther Onkol*. 2007;183 Spec No 2:11–3.

15. Fakiris AJ, Lo SS, Henderson MA, Witt TC, Worth RM, Danis RP, Des Rosiers PM, Timmerman RD. Gamma-knife-based stereotactic radiosurgery for uveal melanoma. *Stereotact Funct Neurosurg.* 2007;85:106–12.
16. Georgopoulos M, Zehetmayer M, Ruhswurm I, Toma-Bstaendig S, Segur-Eltz N, Sacu S, Menapace R. Tumour regression of uveal melanoma after ruthenium-106 brachytherapy or stereotactic radiotherapy with gamma knife or linear accelerator. *Ophthalmologica.* 2003;217:315–9.
17. Hirasawa N, Tsuji H, Ishikawa H, Koyama-Ito H, Kamada T, Mizoe JE, Ito Y, Naganawa S, Ohnishi Y, Tsujii H. Risk factors for neovascular glaucoma after carbon ion radiotherapy of choroidal melanoma using dose-volume histogram analysis. *Int J Radiat Oncol Biol Phys.* 2007;67:538–43.
18. Langmann G, Pendl G, Klaus M, Papaefthymiou G, Guss H. Gamma knife radiosurgery for uveal melanomas: an 8-year experience. *J Neurosurg.* 2000;93 Suppl 3:184–8.
19. Manschot WA, Lee WR, van Strik R. Uveal melanoma: updated considerations on current management modalities. *Int Ophthalmol.* 1995;19:203–9.
20. Marchini G, Gerosa M, Piovan E, Pasoli A, Babighian S, Rigotti M, Rossato M, Bonomi L. Gamma Knife stereotactic radiosurgery for uveal melanoma: clinical results after 2 years. *Stereotact Funct Neurosurg.* 1996;66 Suppl 1:208–13.
21. Muacevic A, Nentwich M, Wowra B, Staerk S, Kampik A, Schaller U. Development of a streamlined, non-invasive robotic radiosurgery method for treatment of uveal melanoma. *Technol Cancer Res Treat.* 2008;7:369–74.
22. Mueller AJ, Schaller U, Talies S, Horstmann GA, Wowra B, Kampik A. Stereotactic radiosurgery using the gamma knife for large uveal melanomas. *Ophthalmologe.* 2003;100:122–8.
23. Mueller AJ, Talies S, Schaller UC, Horstmann G, Wowra B, Kampik A. Stereotactic radiosurgery of large uveal melanomas with the gamma-knife. *Ophthalmology.* 2000;107:1381–7.
24. Rennie I, Forster D, Kemeny A, Walton L, Kunkler I. The use of single fraction Leksell stereotactic radiosurgery in the treatment of uveal melanoma. *Acta Ophthalmol Scand.* 1996;74:558–62.
25. Shields CL, Shields JA, Gunduz K, Freire JE, Mercado G. Radiation therapy for uveal malignant melanoma. *Ophthalmic Surg Lasers.* 1998;29:397–409.
26. Simonova G, Novotny Jr J, Liscak R, Pilbauer J. Leksell gamma knife treatment of uveal melanoma. *J Neurosurg.* 2002;97:635–9.
27. Zehetmayer M, Kitz K, Menapace R, Ertl A, Heinzl H, Ruhswurm I, Georgopoulos M, Dieckmann K, Potter R. Local tumor control and morbidity after one to three fractions of stereotactic external beam irradiation for uveal melanoma. *Radiother Oncol.* 2000;55:135–44.
28. Pe'er J. Ruthenium brachytherapy. *Dev Ophthalmol.* 2012;49:27–40.
29. Zehetmayer M, Menapace R, Kitz K, Ertl A. Experience with a suction fixation system for stereotactic radiosurgery of intraocular malignancies. *Stereotact Funct Neurosurg.* 1995;64 Suppl 1:80–6.
30. Zehetmayer M, Menapace R, Kitz K, Ertl A, Strenn K, Ruhswurm I. Stereotactic irradiation of uveal melanoma with the Leksell gamma unit. *Front Radiat Ther Oncol.* 1997;30:47–55.
31. Muacevic A, Staehler M, Drexler C, Wowra B, Reiser M, Tonn JC. Technical description, phantom accuracy, and clinical feasibility for fiducial-free frameless real-time image-guided spinal radiosurgery. *J Neurosurg Spine.* 2006;5:303–12.
32. Zorlu F, Selek U, Kiratli H. Initial results of fractionated CyberKnife radiosurgery for uveal melanoma. *J Neurooncol.* 2009;94:111–7.